

VPC & Peering

Connecting isolated AWS networks with VPC peering

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00 VPC Recap

A **Virtual Private Cloud (VPC)** is your own isolated network inside AWS. It provides the network environment where resources such as EC2 instances and RDS databases can run and communicate.

The pieces that work together

A VPC is not a firewall. Instead, several networking components work together:

Component	What it does
VPC	Defines the private network and its IP address range.
Subnet	Divides the VPC into smaller networks. Each subnet belongs to one Availability Zone.
Route table	Determines where network traffic can travel.
Internet Gateway	Provides a path between the VPC and the internet.
Public IP	Gives a resource, such as an EC2 instance, an internet-routable address.
Security Group	Acts as a virtual firewall, controlling allowed inbound and outbound traffic.

For example, a public EC2 web server might live inside a subnet with a route to an Internet Gateway. The instance can have a public IP, while its Security Group allows HTTP traffic on port 80.

The important idea

No single setting provides internet access by itself. The network, routing, addressing and security rules must work together.

A VPC can contain multiple subnets

A VPC provides a larger IP address range that can be divided into smaller subnet ranges. Each subnet then provides private IP addresses for the resources placed inside it.

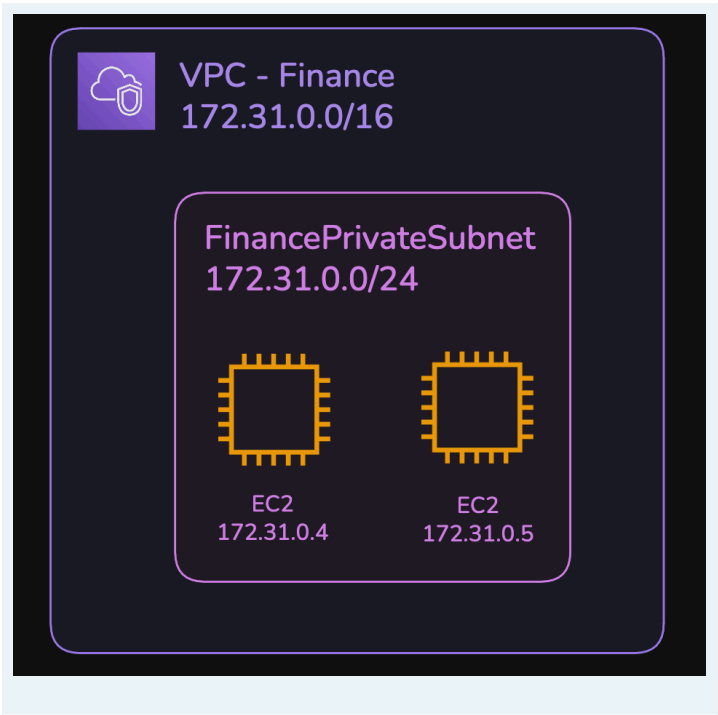
Network	CIDR range
VPC	10.10.0.0/16
Subnet A	10.10.1.0/24
Subnet B	10.10.2.0/24
Subnet C	10.10.3.0/24

A subnet is a range of IP addresses that resources in that subnet can use.

For example, resources inside 10.10.1.0/24 could receive private addresses such as

- 10.10.1.10,
- 10.10.1.11 and
- 10.10.1.12.

This means **multiple EC2 instances can run within the same subnet**. For example, we could create separate private subnets for business areas within an organisation such as **Finance** and **Developers**, keeping each group within its own network area.



Subnets and route tables

Each subnet is associated with a route table. Multiple subnets can share the same route table when they need the same routing behaviour.

Private subnet

Destination	Target
10.10.0.0/16	local

This route allows traffic to stay within the VPC, but there is no direct route to an Internet Gateway.
We call this a private subnet.

Public subnet

Destination	Target
10.10.0.0/16	local
0.0.0.0/0	Internet Gateway

The second route provides a direct path to an Internet Gateway.
A subnet with this routing configuration is considered a public subnet.

How VPC networking fits together

A VPC is the larger private network. Subnets divide its IP range into smaller networks, and resources such as EC2 instances receive private IP addresses from the subnet they belong to.

Each subnet uses a route table to determine where traffic can be sent. For example:

Destination	Target	Meaning
10.10.0.0/16	local	Traffic within the VPC stays within the VPC.
0.0.0.0/0	Internet Gateway	Other IPv4 traffic has a path toward the internet.

0.0.0.0/0 represents all IPv4 destinations.

When its target is an Internet Gateway, the subnet has a route toward the internet. However, that route alone does not give an EC2 instance direct IPv4 internet connectivity.

The EC2 instance also needs a public IPv4 address. Its private IP is used inside the VPC, while the public IP provides the internet-routable address needed for direct external communication.

Route table: determines where the subnet's traffic can be routed.

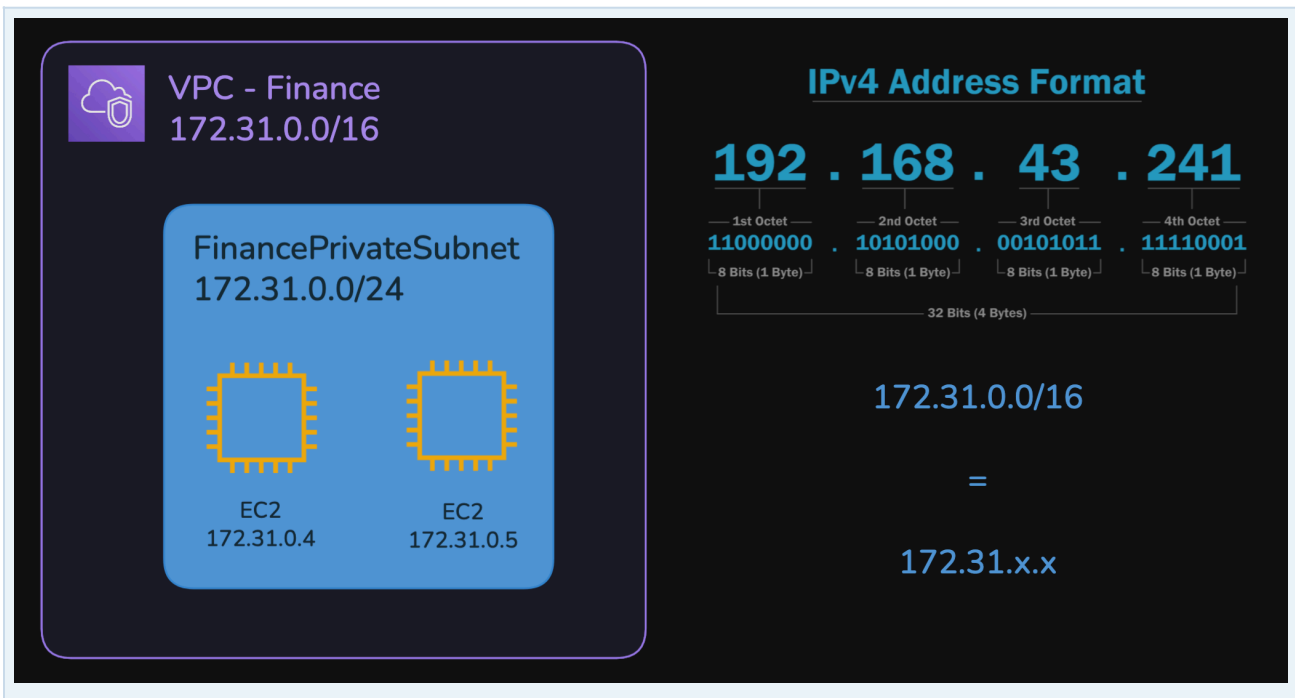
EC2 addressing: determines whether the individual instance has the public address needed to communicate directly with the internet.

This means an EC2 instance can have only a private IP address while still being located inside a public subnet. The subnet is public because of its routing configuration, not because every resource inside it has a public IP.

In short: the VPC provides the network, subnets divide that network, route tables provide routing instructions, and EC2 instances live inside the subnets and use those routes when communicating.

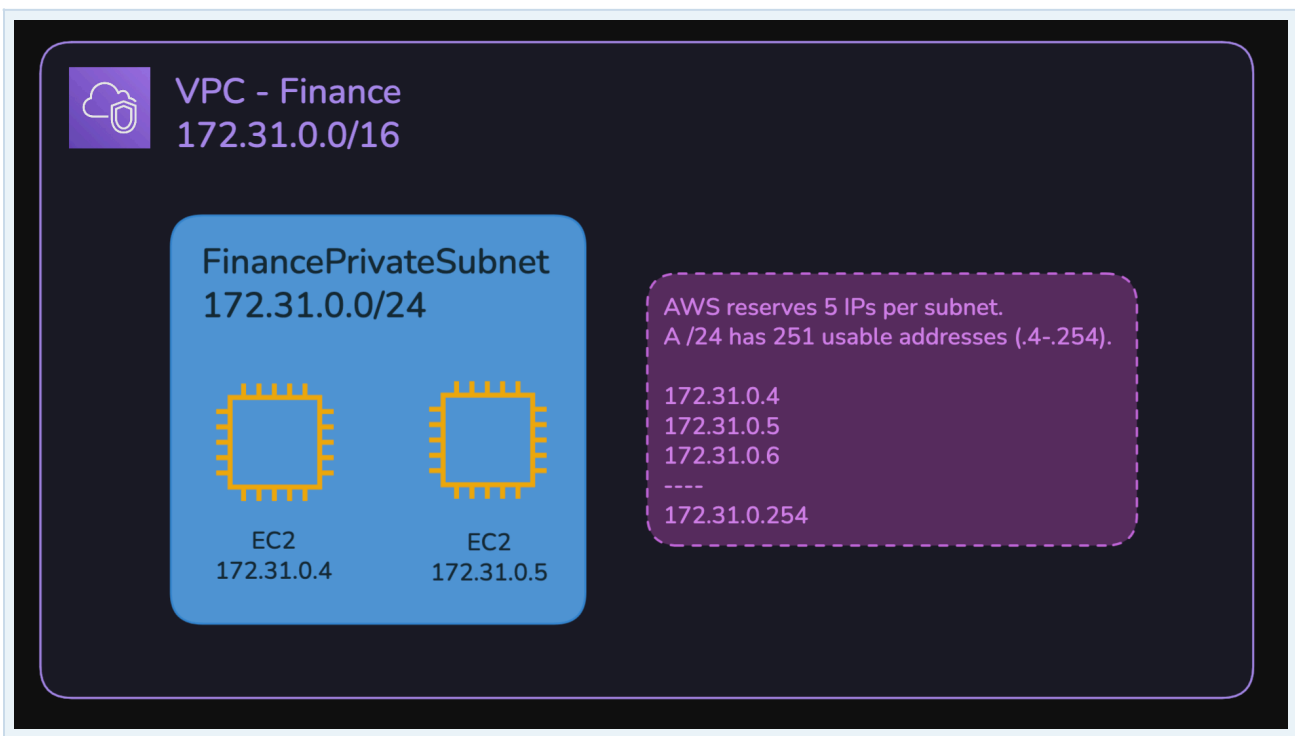
VPC & CIDR Size

A VPC is given a range of private IP addresses using CIDR notation. A /16 fixes the first two octets, leaving the last two available for addressing. For example, 172.31.0.0/16 covers 172.31.0.0 to 172.31.255.255, giving 65,536 total addresses.



Subnets & Available Addresses

A subnet divides the larger VPC address range into smaller networks. A /24 subnet contains 256 addresses. AWS reserves five addresses in each subnet, leaving 251 usable addresses, such as 172.31.0.4 through 172.31.0.254.



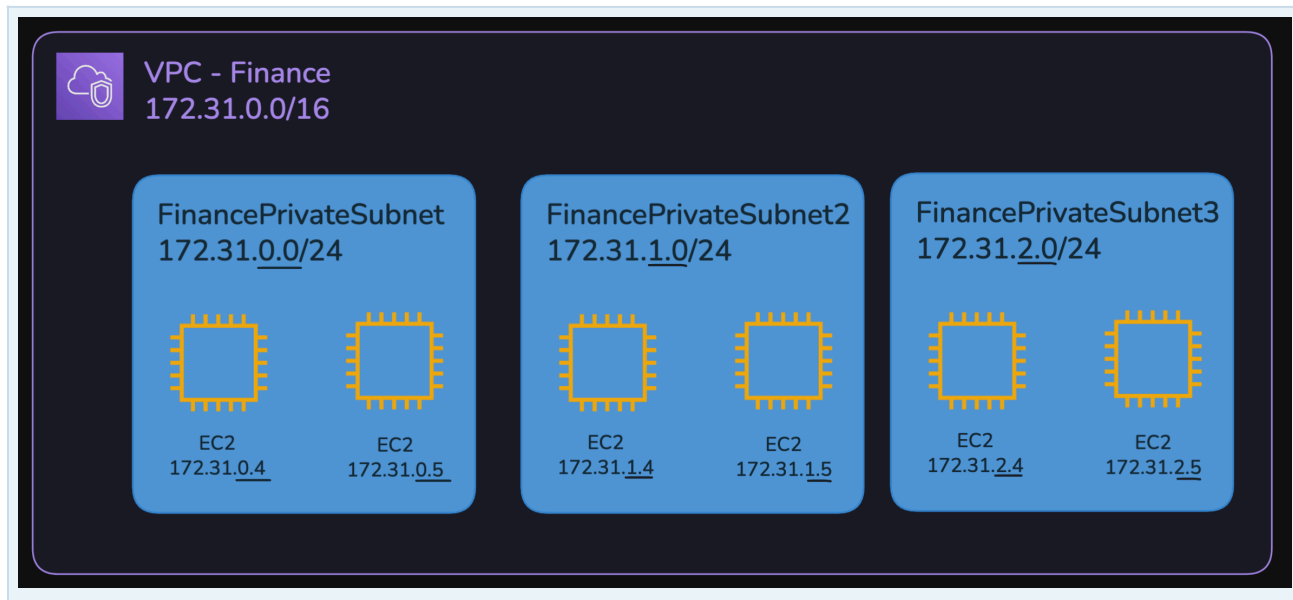
Multiple Subnets

A VPC can contain many subnets, each using a different portion of the VPC address range.
For example,

- 172.31.0.0/24,
- 172.31.1.0/24 and
- 172.31.2.0/24

are all separate networks, but all belong within the larger 172.31.0.0/16 VPC.

Let's visualise it:



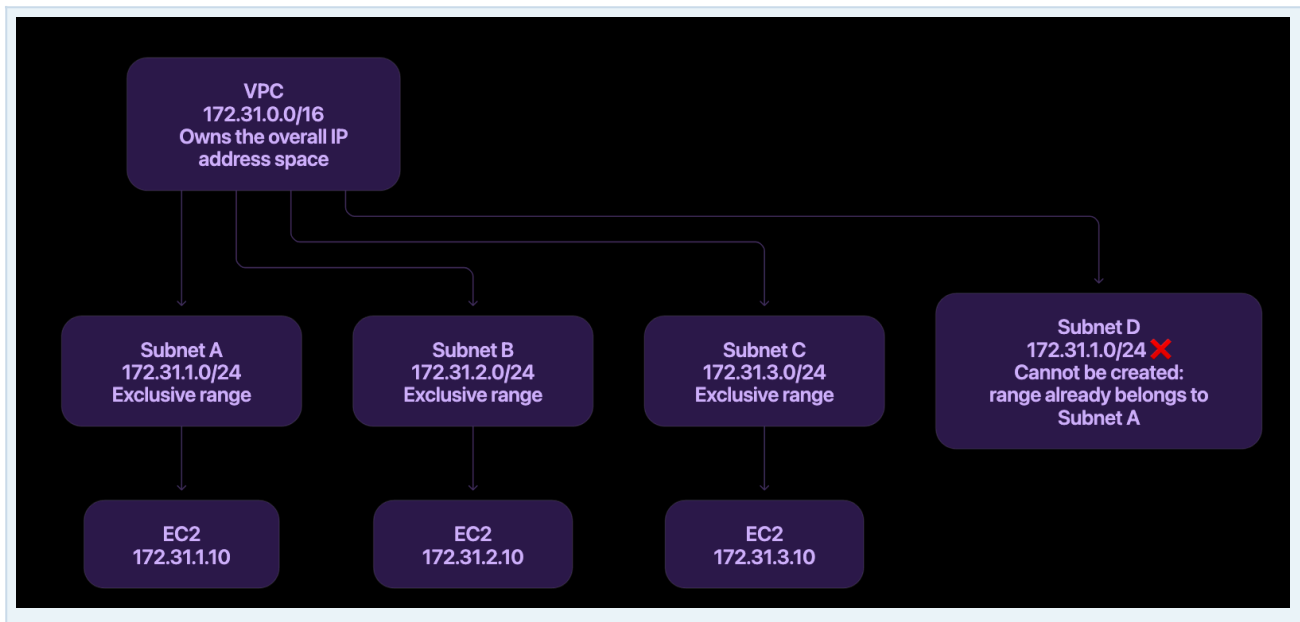
Think about it

If the VPC defines the overall IP range, what does the subnet's CIDR block do?

The VPC defines the **total address space available**. Each subnet is given its **own smaller range within that space**, which is then used by resources placed in that subnet.

That means: an IP address used by a resource in a subnet is also part of the VPC's overall address space.

For example, 172.31.1.10 can belong to a subnet with 172.31.1.0/24, while that entire subnet belongs within the VPC range 172.31.0.0/16



Subnets & Availability Zones

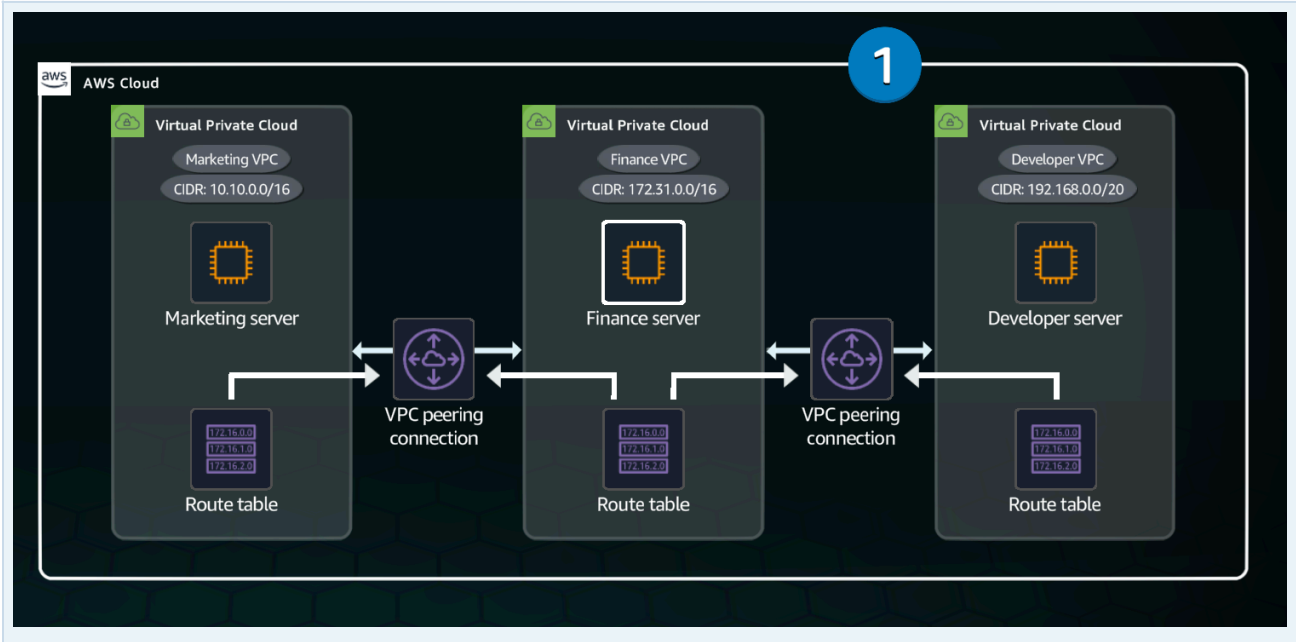
Multiple subnets can exist within the same Availability Zone. However, each individual subnet belongs to only one AZ and cannot span multiple AZs. Creating subnets in different AZs allows resources to be distributed for higher availability.



01 Connecting VPCs

Overview

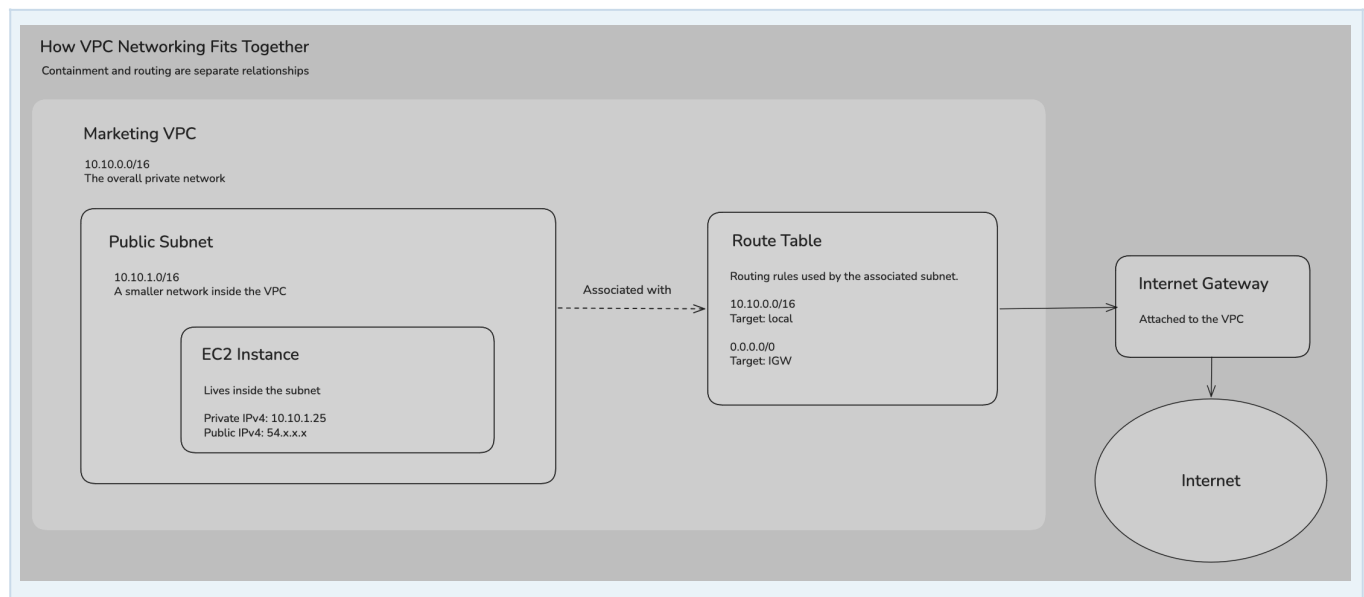
This solution involves three distinct Virtual Private Clouds (VPCs) for the Marketing, Finance and Development departments. The Finance VPC hosts resources required by both the Marketing and Development VPCs.



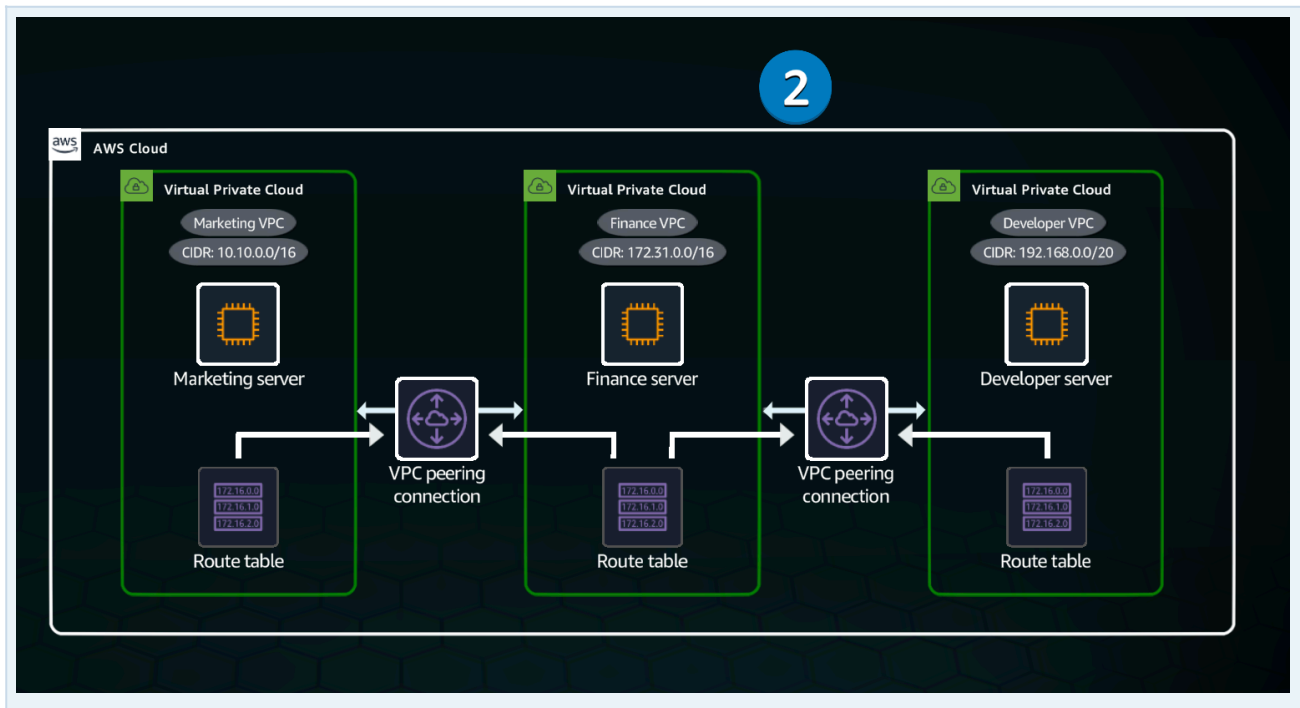
VPCs are isolated by default

Resources in separate VPCs cannot communicate simply because they are in AWS. Access between VPCs requires an explicit network connection.

VPC peering creates a direct private connection between two VPCs through the AWS network. It does not require an Internet Gateway, VPN connection or physical networking hardware.



VPC peering and CIDR blocks



Important: Peered VPCs must use non-overlapping CIDR blocks. Two VPCs that both use 192.168.0.0/24 cannot be peered.

Each VPC has a CIDR block that defines its private IP range. That range can then be divided into smaller ranges for subnets.

VPC	CIDR block
Marketing	10.10.0.0/16
Finance	10.20.0.0/16
Development	10.30.0.0/16

Each VPC can create subnets from its own range. For example, Marketing could use 10.10.1.0/24 and 10.10.2.0/24, while Finance could use 10.20.1.0/24 and 10.20.2.0/24.

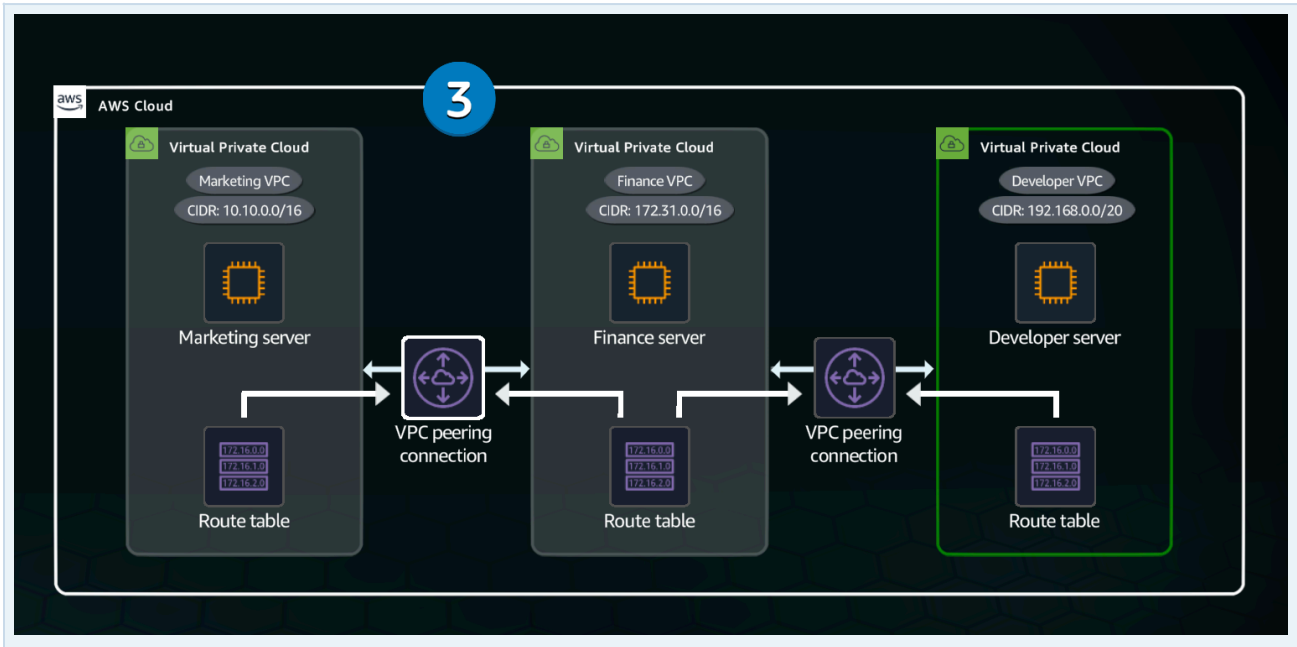
Reusing private IP ranges in completely isolated networks is normally fine. The problem appears when those networks need to communicate: routing must be able to tell which VPC owns the destination address.

For example, Marketing can route 10.10.0.0/16 locally and send 10.20.0.0/16 through the Finance peering connection. Because the ranges are different, AWS knows exactly where the traffic belongs.

Key point: Private IP ranges can be reused across isolated networks, but VPCs connected through peering must use non-overlapping CIDR blocks so traffic can be routed to the correct network.

Marketing connects to Finance

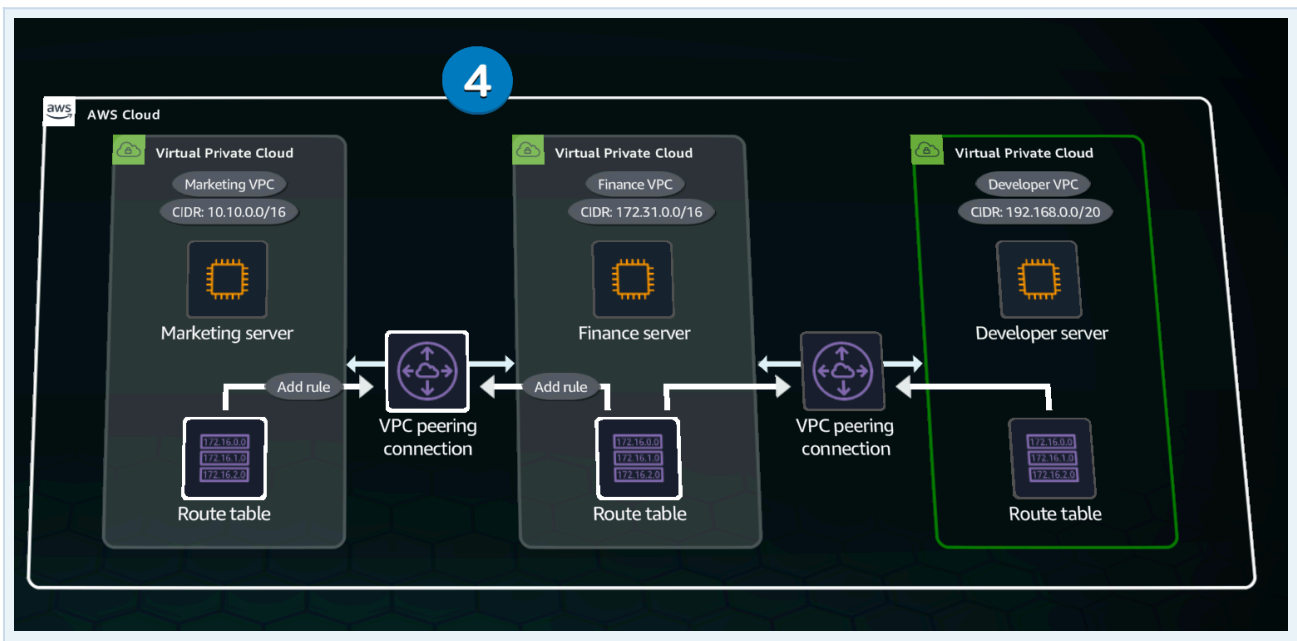
A VPC peering connection is established between the Marketing and Finance VPCs, enabling private IPv4 or IPv6 traffic to be routed between them.



Routing must work in both directions

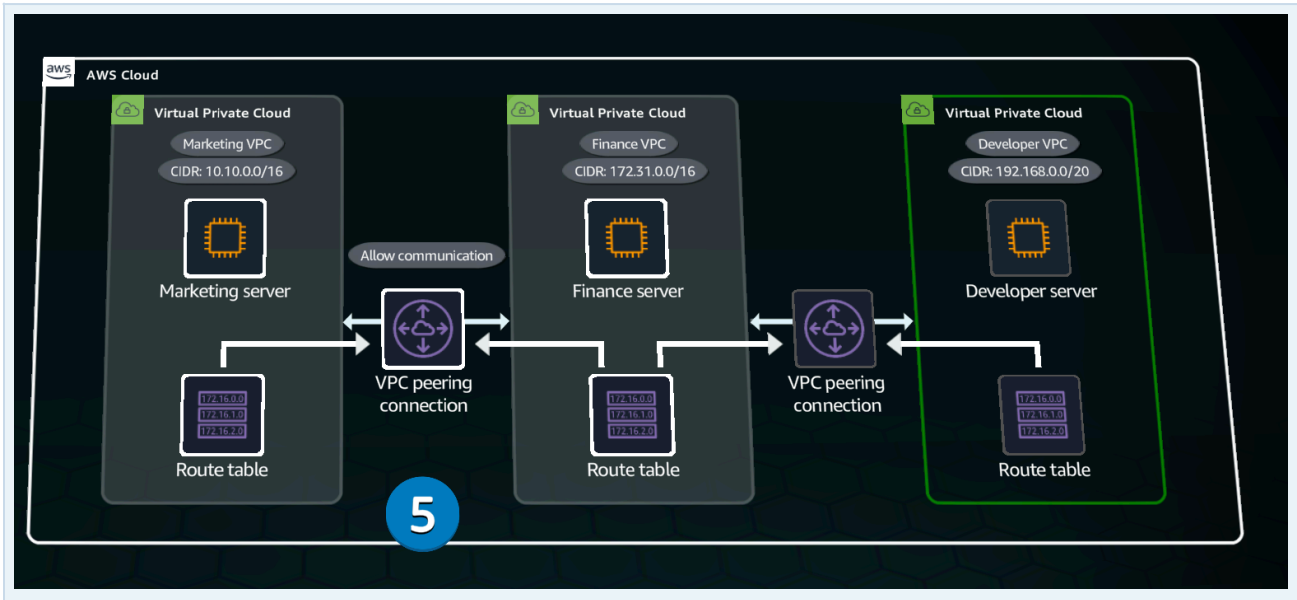
Creating the peering connection is not enough. Route tables in both VPCs need routes that point traffic for the other VPC to the peering connection.

The Marketing route table sends Finance-bound traffic through the peering connection, while the Finance route table provides the return path back to Marketing.



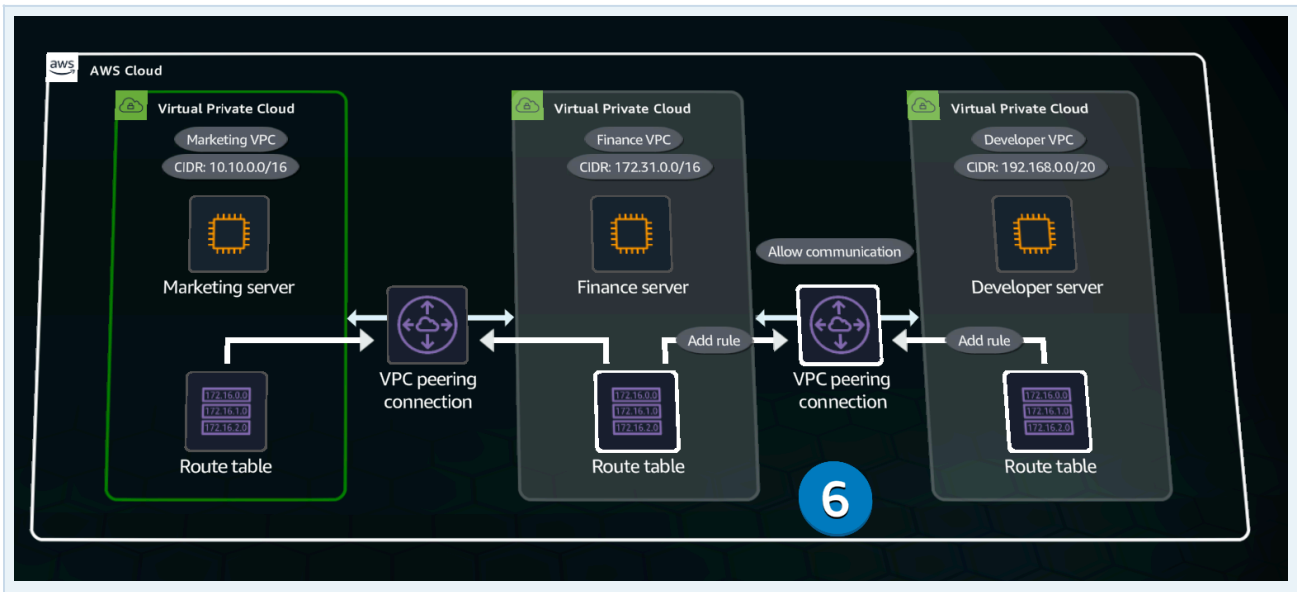
The VPCs can now communicate

Once the peering connection is active and the required routes and security rules are configured, resources in Marketing and Finance can communicate using their private addresses.



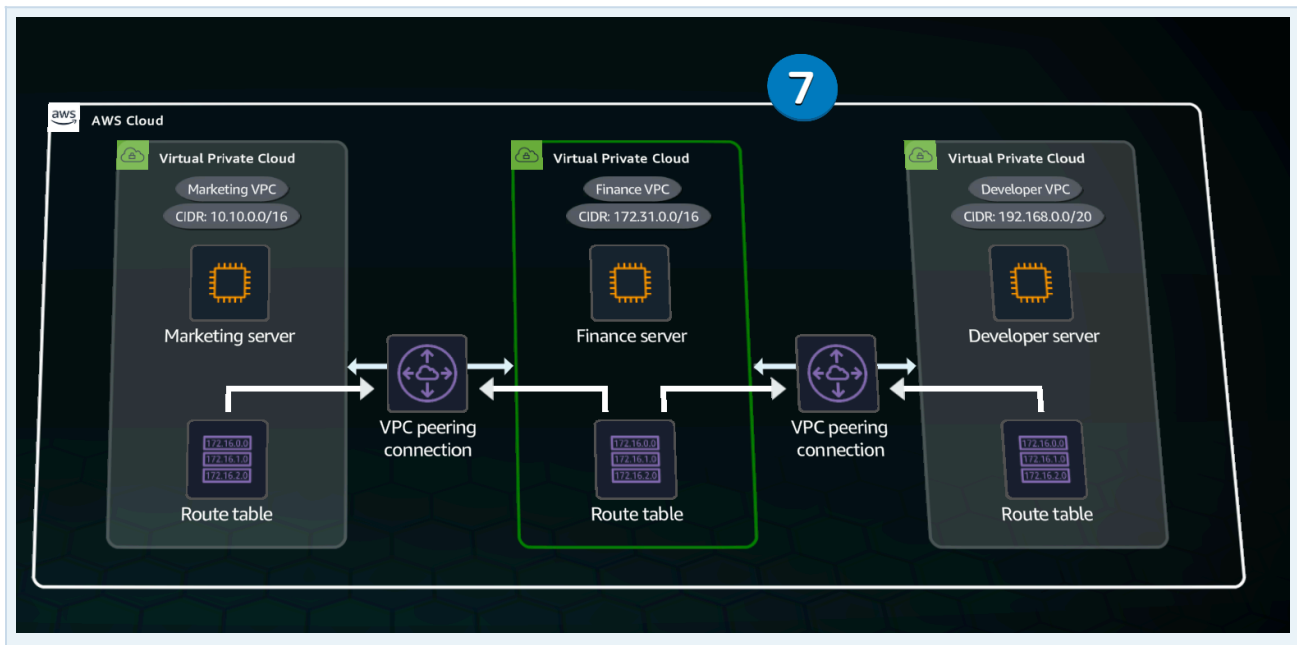
Development also needs Finance

Development requires its own separate peering connection to Finance, with corresponding route table updates in both VPCs.



Why Marketing and Development remain isolated

No peering connection exists between Marketing and Development because they do not require direct communication. VPC peering is not transitive, so traffic cannot travel from Marketing through Finance and then into Development.



Why can't they use the same peering connection?

A VPC peering connection connects exactly two VPCs. It has two endpoints: a requester VPC and an acceptor VPC. Marketing and Finance therefore need one peering connection, while Development and Finance need another.

Those two connections still do not let Marketing and Development communicate through Finance. Finance cannot act as a middleman because VPC peering does not support transitive routing.

When peering becomes difficult to manage

If many VPCs need to connect through a central networking hub, maintaining many individual peering connections can become cumbersome. AWS Transit Gateway is designed for this type of architecture.

02 Connecting VPCs

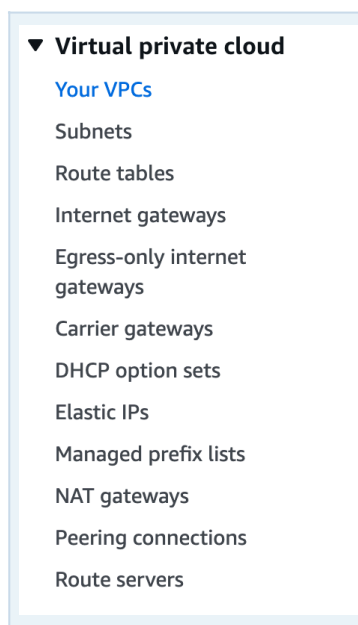
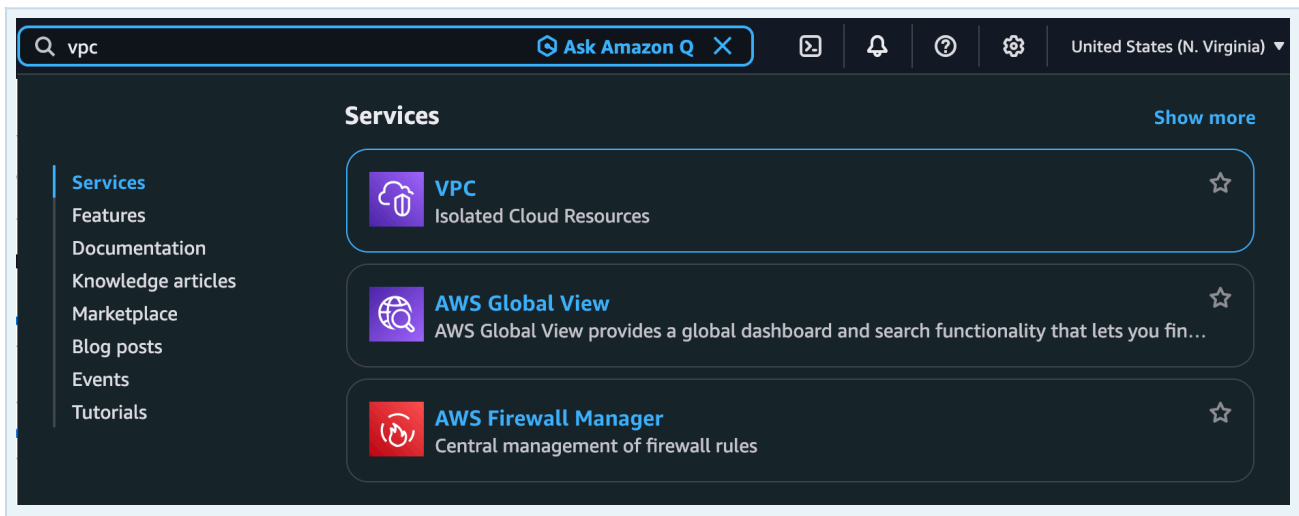
Why can't both VPCs use the same peering connection?

A VPC peering connection connects exactly two VPCs: a requester and an acceptor. Marketing and Finance therefore need one connection, while Development and Finance need another. Finance cannot act as a middleman between them because VPC peering is not transitive. If many VPCs need a central networking hub, AWS Transit Gateway is designed for that type of architecture.

Practice Lab

Goals: set up a VPC peering connection and make sure traffic is properly routed between the peered VPCs.

Amazon Virtual Private Cloud (VPC) lets us launch AWS resources inside a logically isolated virtual network that we define and control.



Here we can see the Finance, Marketing and Development VPC structure described earlier. By default, these VPCs are isolated and cannot communicate with one another.

Your VPCs (4) [Info](#)

Last updated 1 minute ago [Actions](#) [Create VPC](#)

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public..
☐	connecting-vpc/Finance VPC	vpc-01ad9784a1fd9550f	Available	–	–	Off
☐	connecting-vpc/Marketing VPC	vpc-055e476afa49c5d61	Available	–	–	Off
☐	–	vpc-03bd77cbe959170b5	Available	–	–	Off
☐	connecting-vpc/Developer VPC	vpc-04bea4b80702fc5e4	Available	–	–	Off

Verify the EC2 network placement

VPC peering lets resources in two separate VPCs communicate using their private IP addresses. Every EC2 instance runs inside a VPC and a subnet. In the EC2 console, we can verify each instance's VPC, subnet and private IP address and confirm that Marketing, Finance and Development belong to their respective VPCs.

Instances (3) [Info](#)

Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
[Choose filter set](#)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	connecting-vpc/DeveloperServer	i-054239f7ffcd7524	Running	t3.micro	3/3 checks passed	us-east-1a
☐	connecting-vpc/FinanceServer	i-01def3c0f56b3964d	Running	t3.micro	3/3 checks passed	us-east-1a
☐	connecting-vpc/MarketingServer	i-004e2a8db6af3fb96	Running	t3.micro	3/3 checks passed	us-east-1a

A subnet exists entirely within one Availability Zone. In this lab, the Finance instance has no public IP address or public DNS name, confirming that we must reach it through private network connectivity.

☒	connecting-vpc/FinanceServer	i-01def3c0f56b3964d	Running	t3.micro	3/3 checks passed	us-east-1a
☐	connecting-vpc/MarketingServer	i-004e2a8db6af3fb96	Running	t3.micro	3/3 checks passed	us-east-1a

i-01def3c0f56b3964d (connecting-vpc/FinanceServer)

Instance summary [Info](#)

Instance ID i-01def3c0f56b3964d	Public IPv4 address -	Private IPv4 addresses 172.31.0.177
--	--	--

Record the Finance private IP

On the Finance EC2 instance, copy the Private IPv4 address and save it. We will use this address when testing connectivity from Marketing.

Name	Instance ID	State	Instance type	Status check	Availability Zone
connecting-vpc/FinanceServer	i-01def3c0f56b3964d	Running	t3.micro	3/3 checks passed	us-east-1a
connecting-vpc/MarketingServer	i-004e2a8db6af3fb96	Running	t3.micro	3/3 checks passed	us-east-1a

i-01def3c0f56b3964d (connecting-vpc/FinanceServer)

▼ Instance summary Info

Instance ID	Public IPv4 address	Private IPv4 addresses
i-01def3c0f56b3964d	-	172.31.0.177

Testing connectivity: Marketing to Finance

Before configuring peering, we first prove that the two isolated VPCs cannot yet communicate. Select MarketingServer and choose Connect.

Instances (1/3) Info

Last updated 5 minutes ago

Connect Instance state Actions Launch instances

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
connecting-vpc/DeveloperServer	i-054239f7ffcdf7524	Running	t3.micro	3/3 checks passed	us-east-1a
connecting-vpc/FinanceServer	i-01def3c0f56b3964d	Running	t3.micro	3/3 checks passed	us-east-1a
connecting-vpc/MarketingServer	i-004e2a8db6af3fb96	Running	t3.micro	3/3 checks passed	us-east-1a

Session Manager, a capability of AWS Systems Manager, provides browser-based terminal access to EC2 instances. In this lab it lets us access MarketingServer without a public IP address, SSH keys or an open SSH port. Commands entered here run on MarketingServer itself.

In web browser In SSH client

Choose how to connect

- Public subnet access Managed SSH keys
EC2 Instance Connect
Use for instances with public IPs or accessible via standard routing. Requires network connectivity to your instance. Connect using temporary SSH keys managed by AWS.
- Private subnet access Managed SSH keys
EC2 Instance Connect Endpoint
Use for instances in private subnets. Traffic is routed through a VPC endpoint. Connect using temporary SSH keys managed by AWS.
- Maximum security IAM authentication**
SSM Session Manager
Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.
- Direct Troubleshoot
EC2 Serial Console
Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.

Ping the Finance private IP copied earlier. The command receives no replies because there is no route from the Marketing VPC to the Finance VPC yet. Press Ctrl+C to stop the ping.

```
sh-5.2$ ping 172.31.0.177
PING 172.31.0.177 (172.31.0.177) 56(84) bytes of data.
```

Find the Marketing route table

EC2 instances reside inside subnets. From MarketingServer, open its Subnet ID to reach the Marketing subnet.

The screenshot shows the AWS Management Console page for an EC2 instance named 'MarketingServer' (ID: i-004e2a8db6af3fb96). The instance is in the 'Running' state, using a 't3.micro' instance type, and has 3/3 checks passed. It is located in the 'us-east-1a' availability zone. The 'Networking' tab is selected, showing the 'VPC ID' as 'vpc-055e476afa49c5d61' (connecting-vpc/Marketing VPC) and the 'Subnet ID' as 'subnet-04ac3936ec3781632' (connecting-vpc/Marketing VPC/MarketingPublicSubnet1). The 'Availability zone' is 'us-east-1a'.

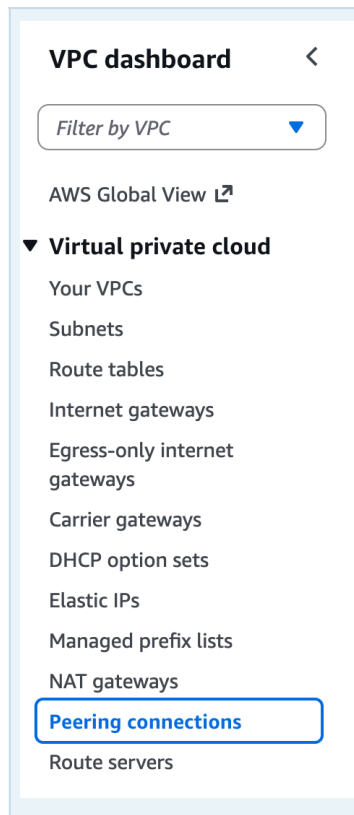
Each subnet is associated with a route table. The route table determines where traffic for different destination networks should be sent.

The screenshot shows the AWS Management Console page for the subnet 'subnet-04ac3936ec3781632' (connecting-vpc/Marketing VPC/MarketingPublicSubnet1). The subnet is in the 'Available' state. The 'Route table' section shows the route table 'rtb-01839311a9e7bbc8d' (connecting-vpc/Marketing VPC/MarketingPublicSubnet1). The route table has two routes: one for destination '10.10.0.0/16' with target 'local', and another for destination '0.0.0.0/0' with target 'igw-00cb4b0a9b2930e63'. The 'Edit route table association' button is visible.

Private VPC networks use private ranges such as 10.x.x.x, 172.16–31.x.x or 192.168.x.x. Routes determine whether traffic stays local, travels to another VPC, or uses a gateway where one is configured.

Create the VPC peering connection

The next step is to create a peering connection between Marketing and Finance.



Choose Create peering connection.

We want a direct private connection between the Marketing and Finance VPCs.



The requester VPC initiates the peering request and the acceptor VPC receives it. Before creating the connection, identify both VPC IDs and their CIDR blocks, and confirm that the CIDR ranges do not overlap.

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Marketing <-> Finance

Select a local VPC to peer with

VPC ID (Requester)
vpc-055e476afa49c5d61 (connecting-vpc/Marketing VPC)

VPC CIDRs for vpc-055e476afa49c5d61 (connecting-vpc/Marketing VPC)

CIDR	Status	Status reason
10.10.0.0/16	✔ Associated	-

Because both VPCs are in the same AWS account, we can select the Finance VPC as the acceptor and manage both sides of the request ourselves.

Select another VPC to peer with

Account
☒ My account
☐ Another account

Region
☒ This Region (us-east-1)
☐ Another Region

VPC ID (Acceptor)
vpc-01ad9784a1fd9550f (connecting-vpc/Finance VPC)

VPC CIDRs for vpc-01ad9784a1fd9550f (connecting-vpc/Finance VPC)

CIDR	Status	Status reason
172.31.0.0/16	✔ Associated	-

Within one account: provide both VPC IDs, create the peering request, then accept it. Finish this step by choosing Create peering connection.

✔ A VPC peering connection **pcx-0417bb17f135d4ac2 / Marketing <-> Finance** has been requested.

pcx-0417bb17f135d4ac2 / Marketing <-> Finance Actions ▲

Pending acceptance
You can accept or reject this peering connection request using the 'Actions' menu. You have until Friday, September 11, 2026 at 09:18:51 GMT+2 the request, otherwise it expires.

Details [Info](#)

Requester owner ID	Acceptor owner ID	VPC Peering connection

- Accept request
- Reject request
- Edit DNS settings
- Manage tags
- Delete peering connection

Across different AWS accounts, the owner of the acceptor VPC must accept the peering request.

Peering accepted - routing is still required

Accepting the request creates the peering connection, but it does not automatically add the routes required for traffic to use it.

✔ Your VPC peering connection (pcx-0417bb17f135d4ac2 | Marketing <=> Finance) has been established. To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables.

Modify my route tables now

Info

pcx is the identifier prefix for a VPC peering connection. Take note of the peering connection ID because we will use it as a route target.

Review the connection and verify that its status is Active before continuing.

pcx-0417bb17f135d4ac2 / Marketing <=> Finance

Actions

Details

Requester owner ID
408687975267

Peering connection ID
pcx-0417bb17f135d4ac2

Status
Active

Expiration time
-

Accepter owner ID
408687975267

Requester VPC
vpc-055e476afa49c5d61 / connecting-vpc/Marketing VPC

Requester CIDRs
10.10.0.0/16

Requester Region
N. Virginia (us-east-1)

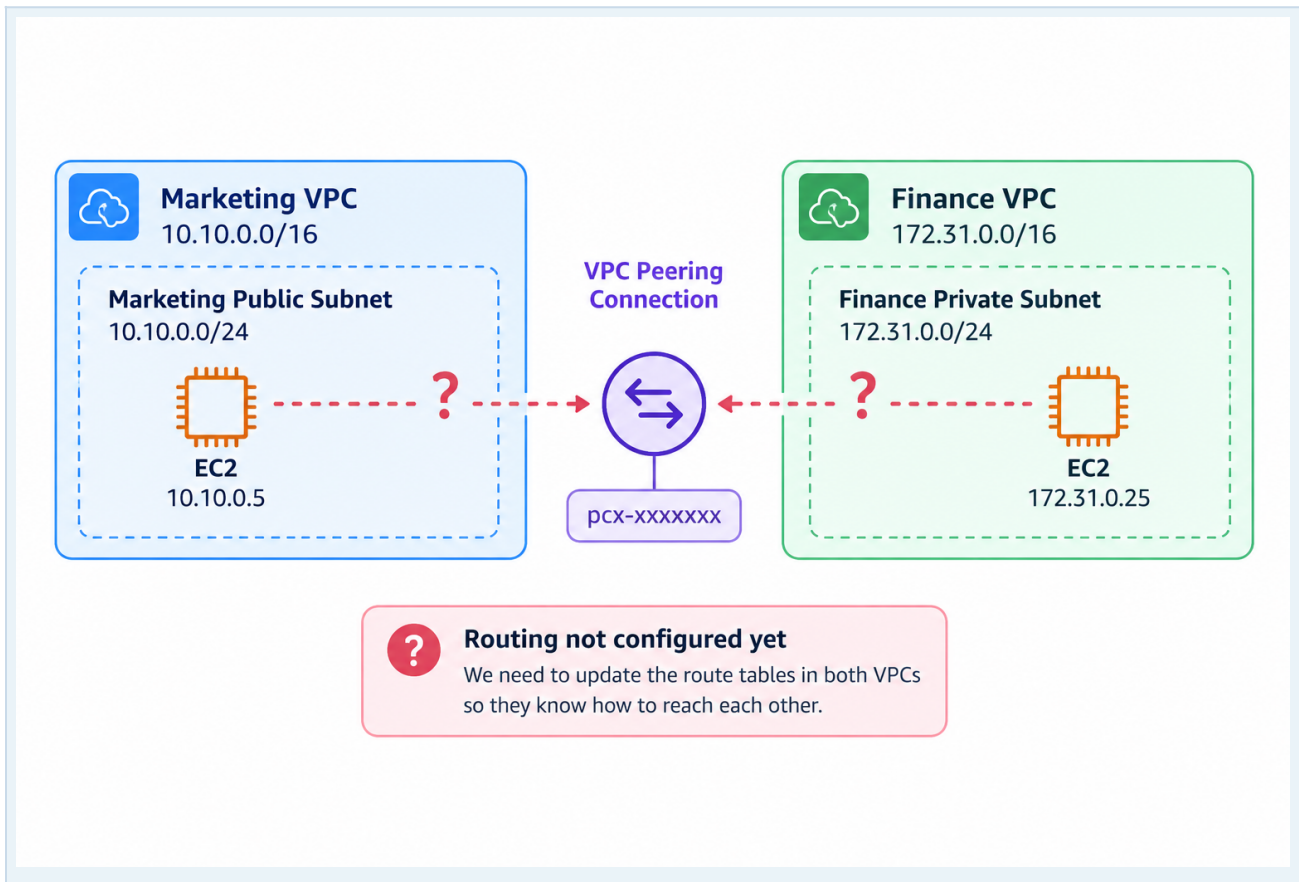
VPC Peering connection ARN
arn:aws:ec2:us-east-1:408687975267:vpc-peering-connection/pcx-0417bb17f135d4ac2

Accepter VPC
vpc-01ad9784a1fd9550f / connecting-vpc/Finance VPC

Accepter CIDRs
172.31.0.0/16

Accepter Region
N. Virginia (us-east-1)

Each VPC still needs a route telling it to use the peering connection when the destination belongs to the other VPC's CIDR block.



Think of the peering connection as a new road between two cities. The road exists, but the route tables provide the directions that tell traffic when to use it.

Update both route tables

Return to the Marketing subnet and open its route table. Add a route for the Finance VPC CIDR block with the VPC peering connection as the target.

connecting-vpc/MarketingServer
i-004e2a8db6af3fb96

Running

t3.micro

3/3 checks passed
us-east-1a

i-004e2a8db6af3fb96 (connecting-vpc/MarketingServer)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

VPC ID
 vpc-055e476afa49c5d61 (connecting-vpc/Marketing VPC)

Subnet ID
 subnet-04ac3936ec3781632 (connecting-vpc/Marketing VPC/MarketingPublicSubnet1)

Availability zone
 us-east-1a

Subnets (1/1) [Info](#)

Last updated 2 minutes ago

Actions

Create subnet

Subnet ID : subnet-04ac3936ec3781632

Clear filters

☒	Name	Subnet ID	State	VPC	Block Public...	IP
☒	connecting-vpc/Marketing VPC/Marketi...	subnet-04ac3936ec3781632	Available	vpc-055e476afa49c5d61 conn...	Off	10

subnet-04ac3936ec3781632 / connecting-vpc/Marketing VPC/MarketingPublicSubnet1

Availability Zone

 use1-az4 (us-east-1a)

Network ACL

[acl-0a8eaeccfd4aa814a](#)

Auto-assign customer-owned IPv4 address

No

Available IPv4 addresses

 249

Network border group

 us-east-1

Default subnet

No

Customer-owned IPv4 pool

-

VPC

[vpc-055e476afa49c5d61](#) | [connecting-vpc/Marketing VPC](#)

Auto-assign public IPv4 address

Yes

Outpost ID

-

Route table

[rtb-01839311a9e7bbc8d](#) | [connecting-vpc/Marketing VPC/MarketingPublicSubnet1](#)

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Allow the traffic in the Security Group

Routing provides the network path, but the relevant Security Group must also allow the traffic being tested. Update the Finance security rules as required by the lab.

sg-09f19d5b955406085 - FinanceServerSecurityGroup Actions

Details

Security group name
FinanceServerSecurityGroup

Security group ID
sg-09f19d5b955406085

Description
Security Group for Finance Server

VPC ID
[vpc-0aefe2490392b9ee0](#)

Owner
411181719368

Inbound rules count
0 Permission entries

Outbound rules count
1 Permission entry

Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules Manage tags Edit inbound rules

Name

Security group rule ID

IP version

Type

Protocol

Port range

No security group rules found

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-02f3f2431a93453e9	All ICMP - IPv4	ICMP	All	Custom 192.168.0.0/20		Delete
sgr-050f2dac07c7208e5	All ICMP - IPv4	ICMP	All	Custom 10.10.0.0/16		Delete

Add rule

Cancel Preview changes Save rules

Test again

Return to MarketingServer through Session Manager and ping the Finance private IP again. With the peering connection, routes and security rules in place, the FinanceServer now responds.

```
sh-5.2$ ping 172.31.0.117
PING 172.31.0.117 (172.31.0.117) 56(84) bytes of data.
64 bytes from 172.31.0.117: icmp_seq=1 ttl=127 time=0.271 ms
64 bytes from 172.31.0.117: icmp_seq=2 ttl=127 time=0.288 ms
64 bytes from 172.31.0.117: icmp_seq=3 ttl=127 time=0.278 ms
^C
--- 172.31.0.117 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2061ms
rtt min/avg/max/mdev = 0.271/0.279/0.288/0.007 ms
sh-5.2$
```

Success!

The two VPCs can now communicate privately through the VPC peering connection.

Summary

A **VPC** is an isolated private network in AWS.

It defines the overall IP address space available to resources inside it.

A VPC can contain **multiple subnets**.

Each subnet receives its own non-overlapping CIDR range from within the VPC's larger range, and resources such as EC2 instances receive **private IP addresses** from the subnet they belong to.

Route tables determine where traffic can travel.

A subnet with a route to an Internet Gateway can provide a path toward the internet, while private networks can remain without direct internet access.

Separate VPCs are **isolated by default**.

When resources in two VPCs need to communicate privately, **VPC peering** can create a direct connection between them. Both sides still need the correct routes and security rules before traffic can flow.

Remember

VPC - Defines the overall private network and IP range

Subnet - Defines a smaller IP range within the VPC

EC2 private IP - Comes from the subnet's range

Route Table - Determines where traffic can go

VPC Peering - Connects two VPCs privately

Security Group - Determines whether traffic is allowed

The path must exist, and the traffic must be allowed.